AZIZ, Karim

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EDUCATION

Boston University, Boston, MA, U.S.A. — September 2023 - May 2027 (Expected)

Bachelor of Arts in Mathematics (Pure Track), Bachelor of Arts in Physics (Graduate Option)

- Grade Point Average: 3.93/4.00
- Relevant Coursework: Honors Vector Calculus, Honors Differential Equations, Honors Linear Algebra, Discrete Mathematics, Abstract Algebra I & II, Real Analysis I & II, Topology, Algebraic Geometry, Differntial Geometry, Algebraic Number Theory, Classical Mechanics, Modern Physics, Electrodynamics I & II, Quantum Physics I & II, Introduction to Computer Science, Research Writing

Collège de la Sainte Famille (Jesuits), Cairo, Egypt — September 2019 - July 2023

French Baccalauréat with Mathematics and Physics-Chemistry Specialty

- Grade Average: 18.26/20.00
- Relevant Coursework: Mathematics, Physics, Chemistry, Programming, Elementary Statistics

EXPERIENCE

European Organization for Nuclear Research (CERN), Geneva, Switzerland — January 2026 - Present

Boston University, Boston, MA, U.S.A. — April 2025 - Present

Undergraduate Researcher, Supervisor: Prof. Anatoli Polkovnikov

- Investigated an effective model for classical many-body systems modeling phonon-electron interactions.
- Solved for analytic solutions of these systems and wrote computational simulations to verify them.
- Currently working on solving systems with increasingly larger numbers of vibrational modes.

Boston University, Boston, MA, U.S.A.

Research Experience for Undergraduates "Ambassador" — June - August 2025

- Supervised a group of students who did NSF-funded research at Boston University while helping them discover what Boston has to offer, whether academic or cultural.

RESEARCH

American Physical Society March Meeting, Denver, CO, U.S.A. — March 2026

Oral Presentation

- Presented research conducted under Prof. Polkovnikov at APS's anual conference.

Boston University, Boston, MA, U.S.A. — August 2025

Poster Presentation

- Presented a poster titled "Corrections to the Born-Oppenheimer Approximation in Classical Systems" on research conducted over the summer in theoretical condensed matter physics. This investigated how a result used in quantum mechanics could be translated to classical systems modeling phonon-electron interactions.

GRANTS AND SCHOLARSHIPS

Boston University, Boston, MA, U.S.A. — September 2023 - Present

Full-Ride "University Scholarship"

- Received a scholarship which covers four years of tuition and living expenses. This scholarship is awarded to students "with exceptional financial need and outstanding academic achievement and promise."

Boston University, Boston, MA, U.S.A. — June - August 2025

NSF Grant Number PHY-2244795

- Conducted research over the summer through Boston University's Research Experience for Undergraduates.

SKILLS

Computational: Python | Unix | LaTeX | Wolfram Mathematica
Linguistic: English (*Native*) | French (*Native*) | Arabic (*Native*) | Spanish (*Fluent*)
Extracurricular: Classical Music (*Pianist, Composer*) | Chess (*FIDE Profile*) | Soccer Player (*Right Forward*)